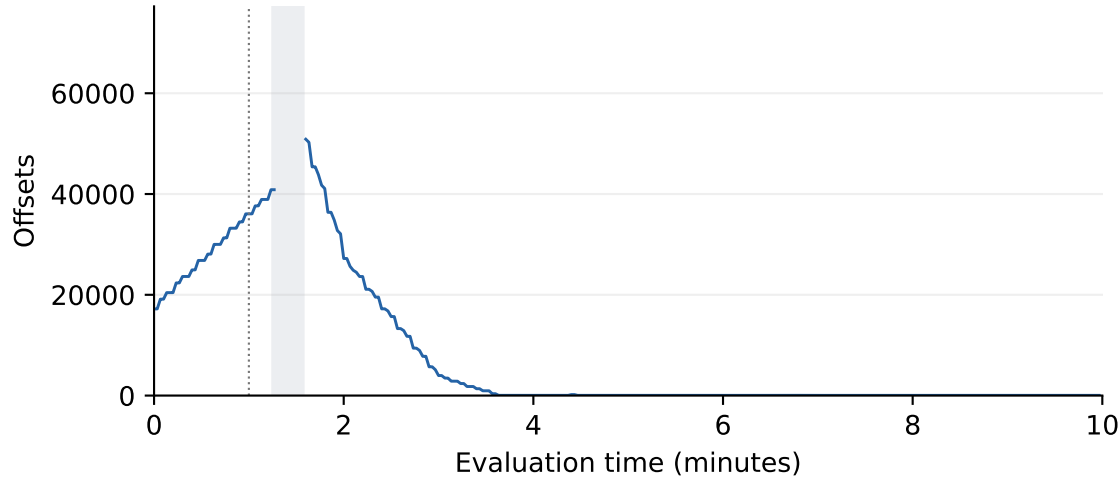
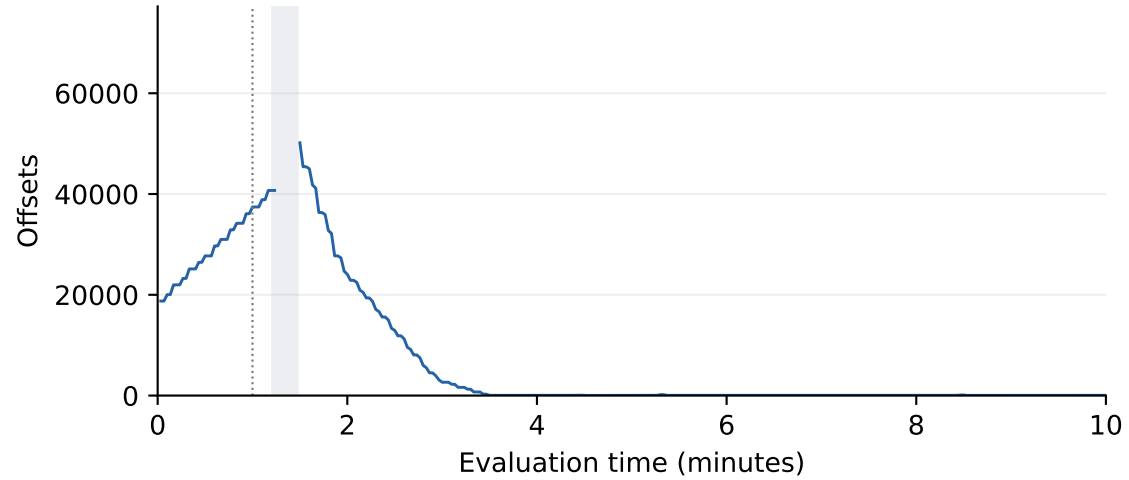


Total lag - 700 messages/s

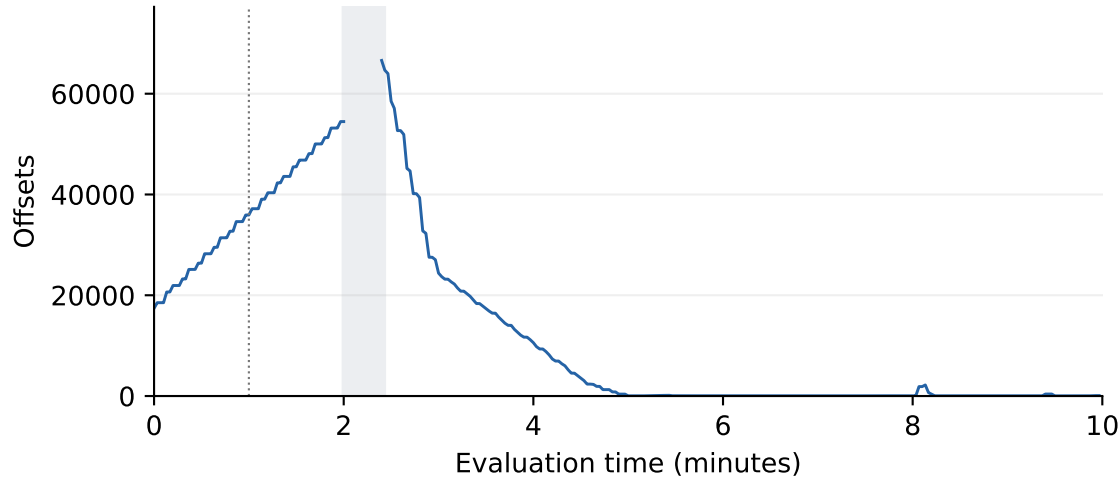
Redistribute within 3 - Run 1



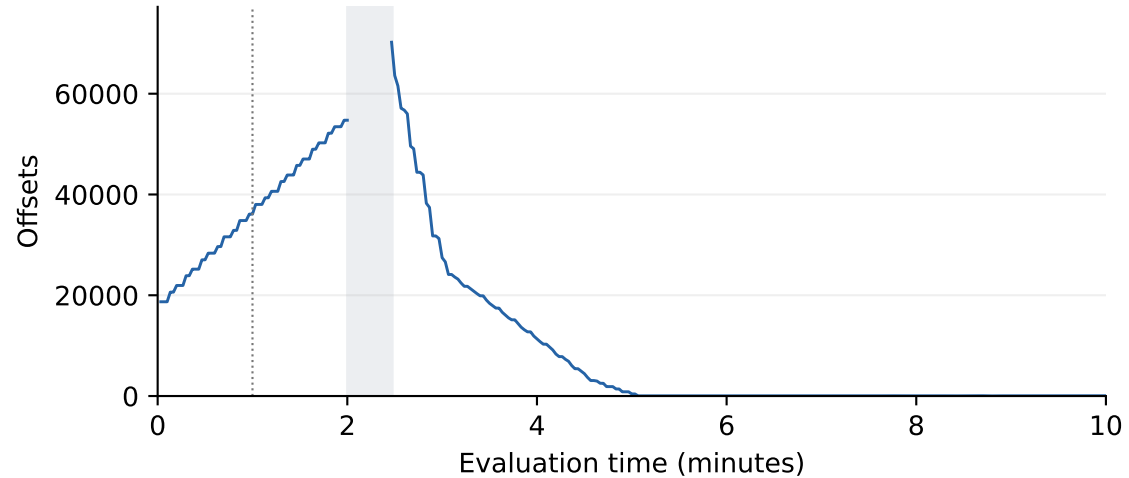
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2

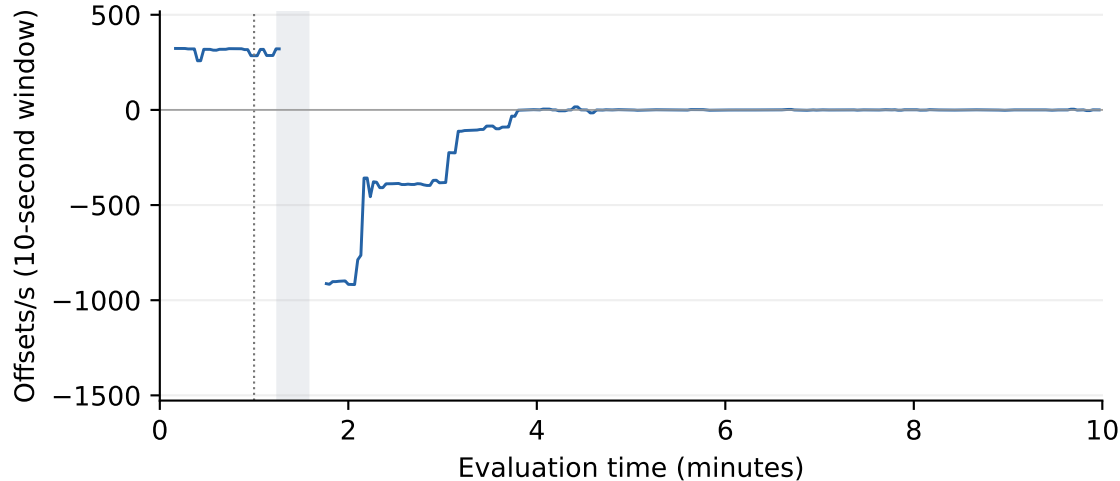


80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

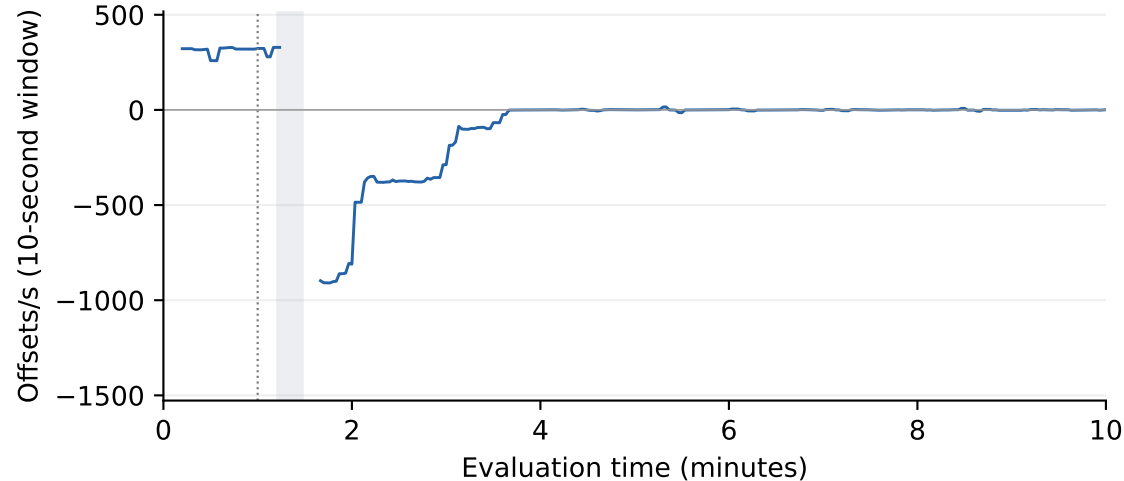
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

Processing-backlog growth - 700 messages/s

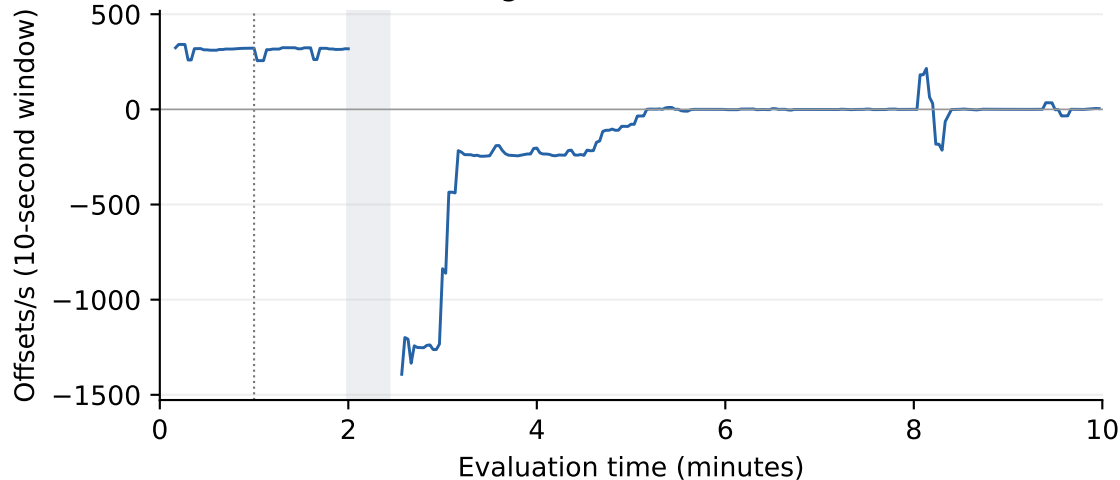
Redistribute within 3 - Run 1



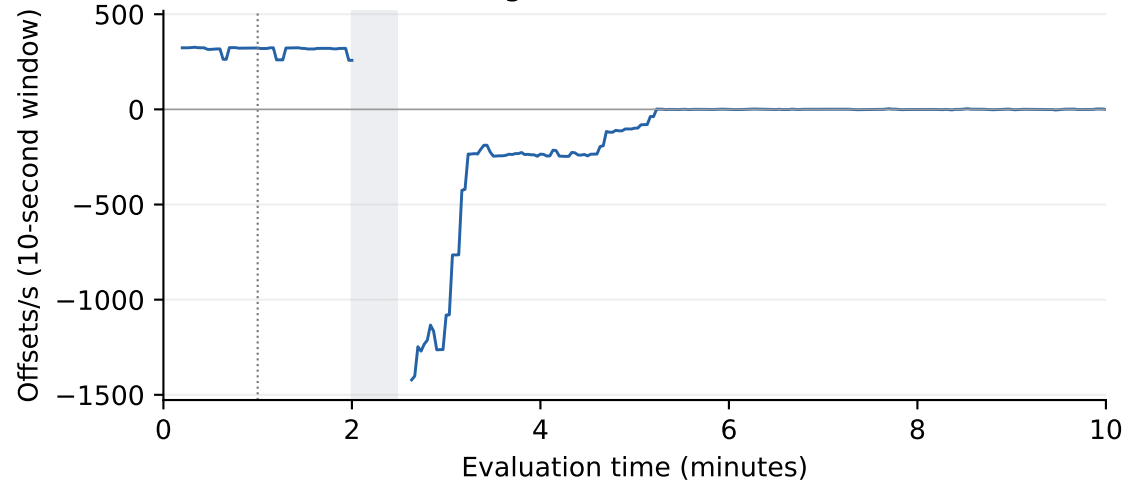
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2



80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

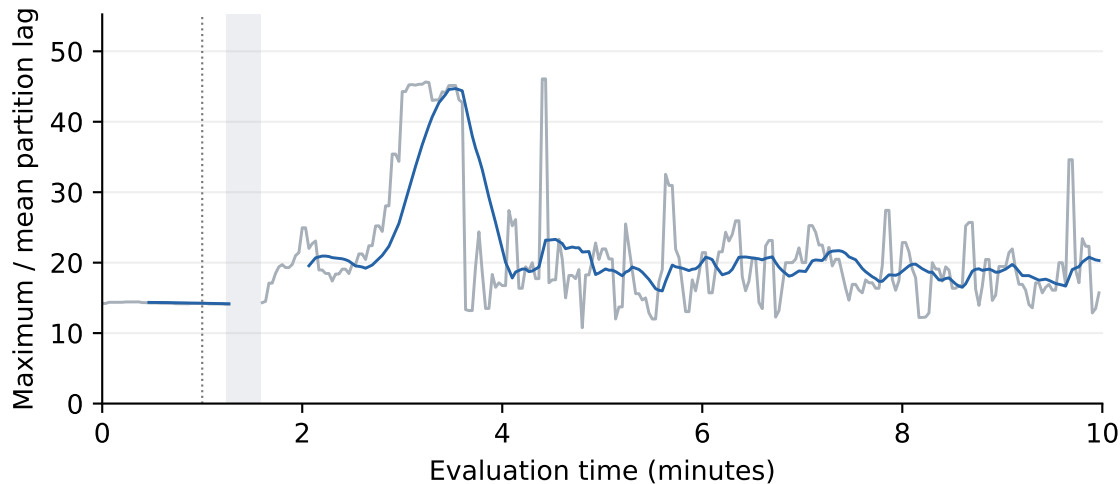
Growth needs 10 seconds of fresh contiguous observations after a break. Dotted line: scheduled intervention.

Shading: recorded transition interval; this differs from the growth-window waiting time.

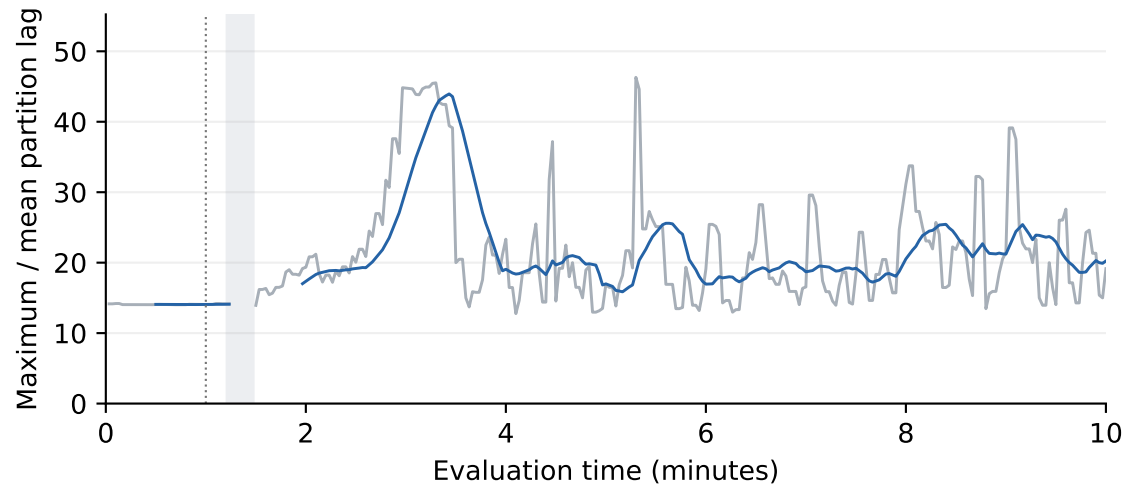
Partition lag skew - 700 messages/s

— Snapshot ratio — 15-sample mean

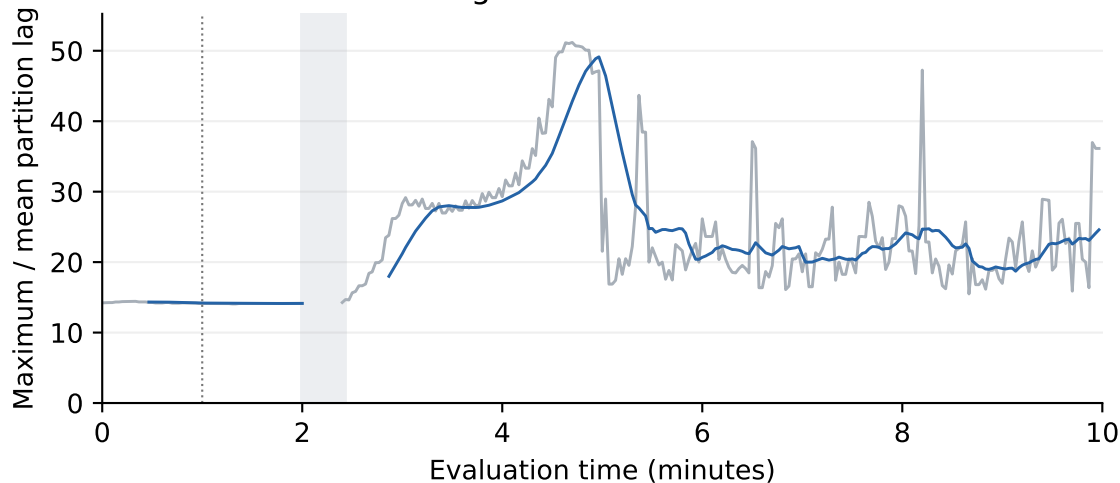
Redistribute within 3 - Run 1



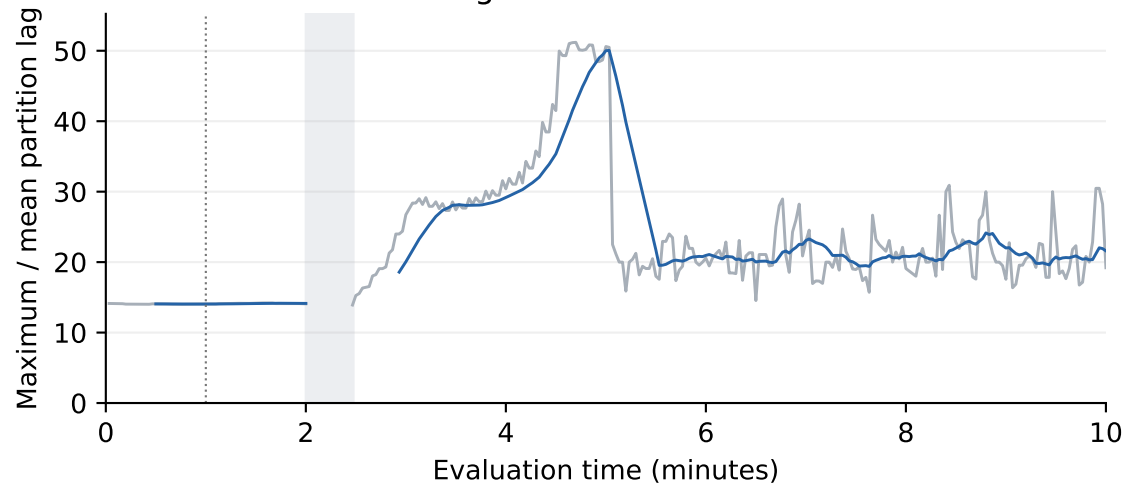
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2



80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

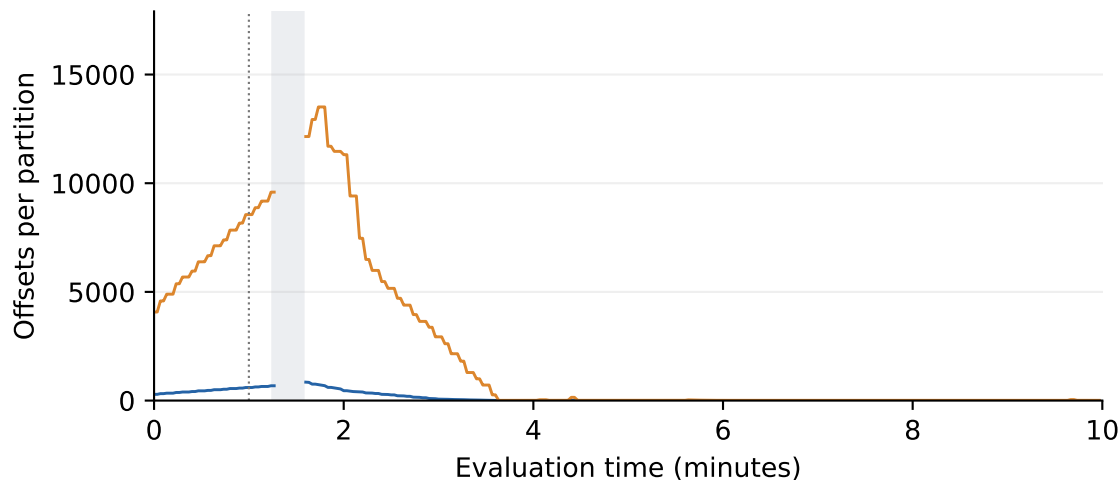
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

Maximum and mean partition lag - 700 messages/s

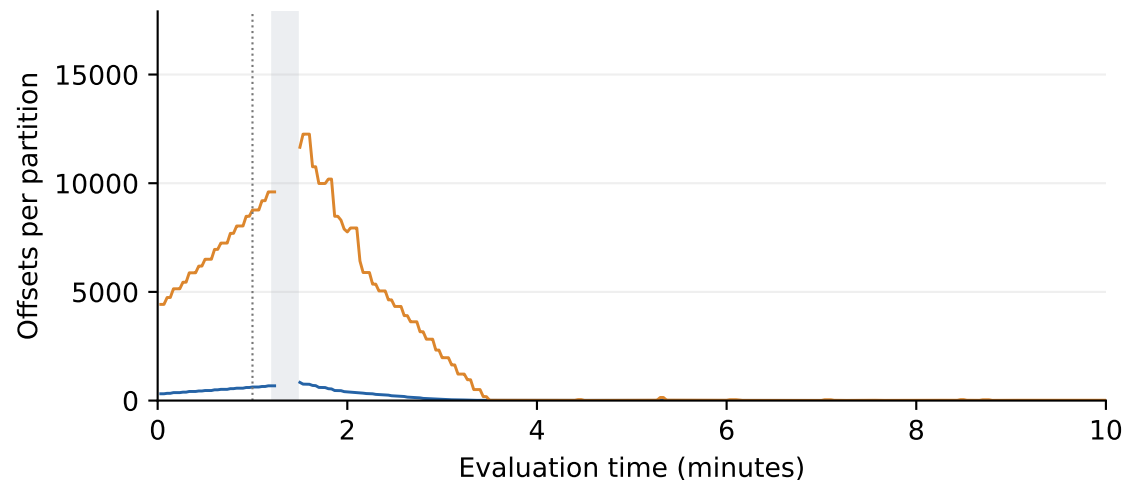
— Mean across partitions

— Maximum partition

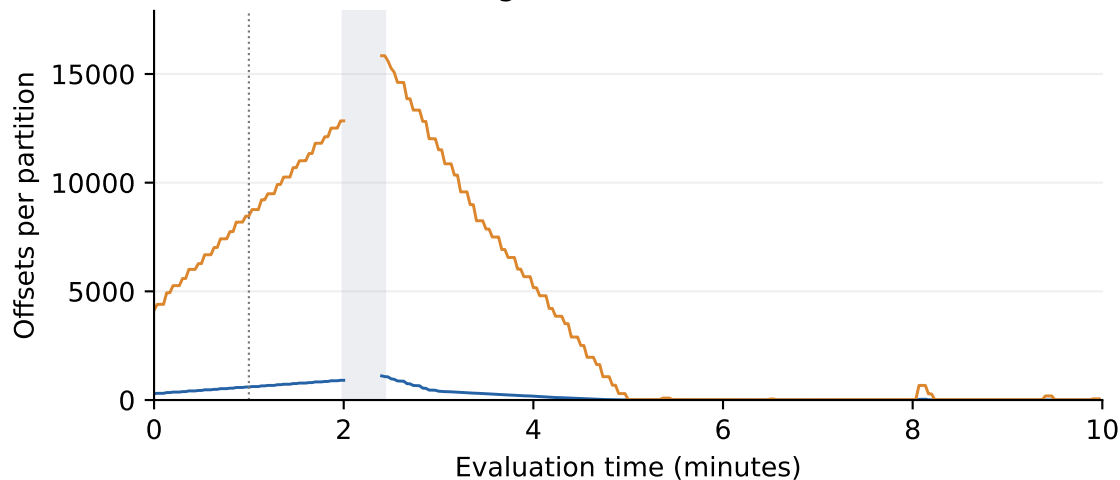
Redistribute within 3 - Run 1



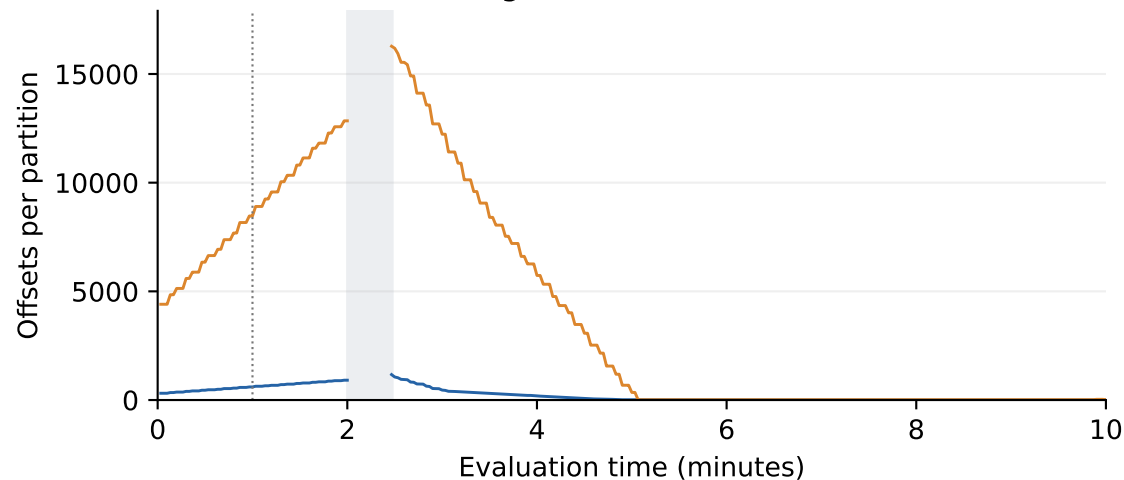
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2

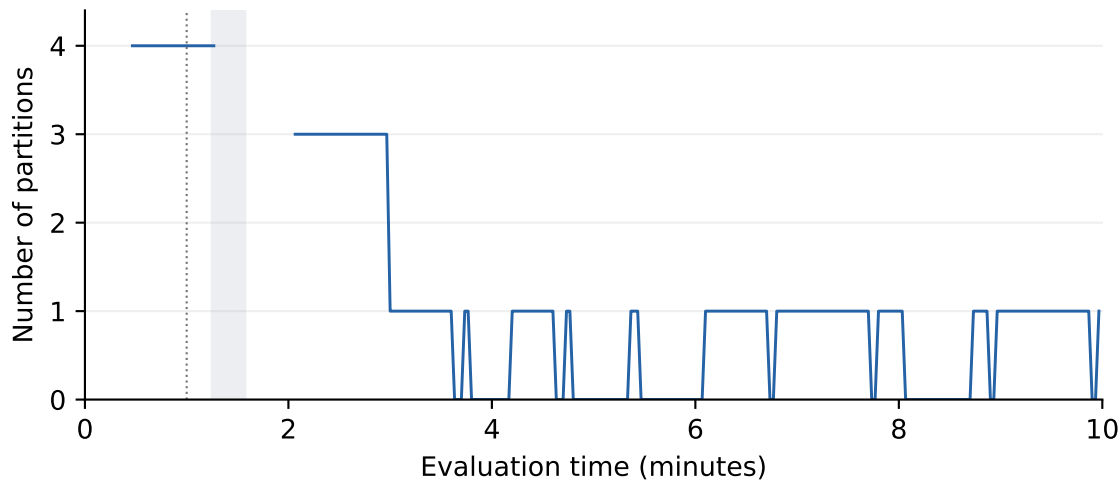


80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

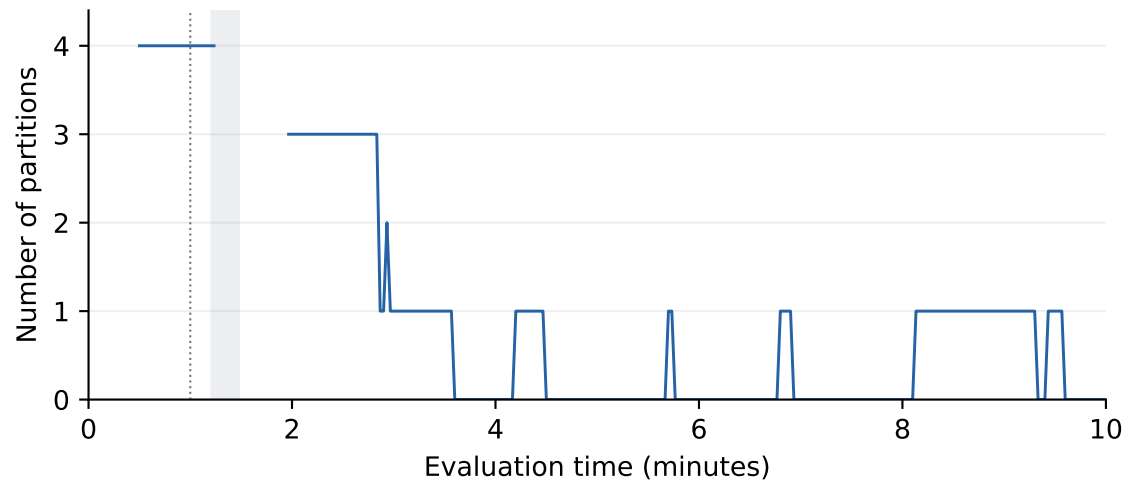
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

Persistently hot partitions - 700 messages/s

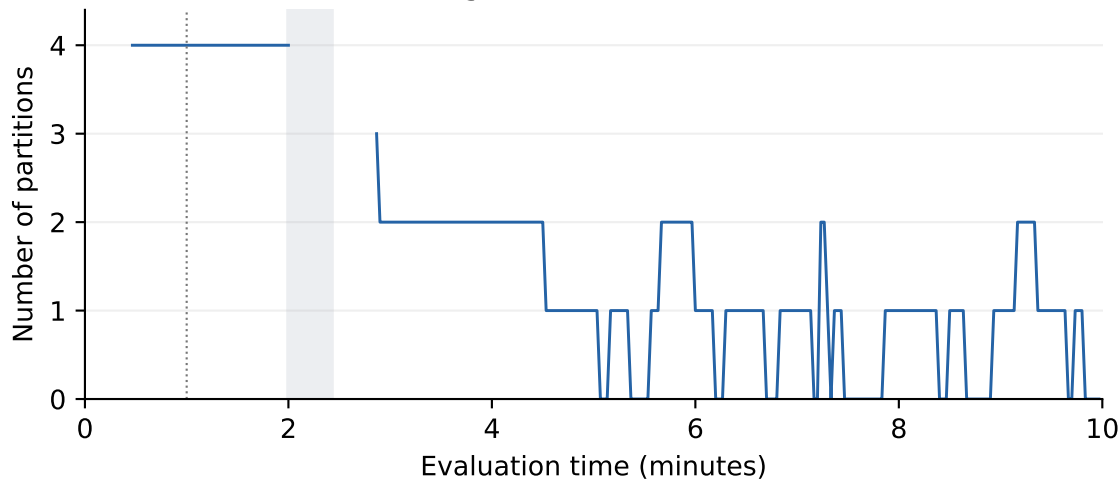
Redistribute within 3 - Run 1



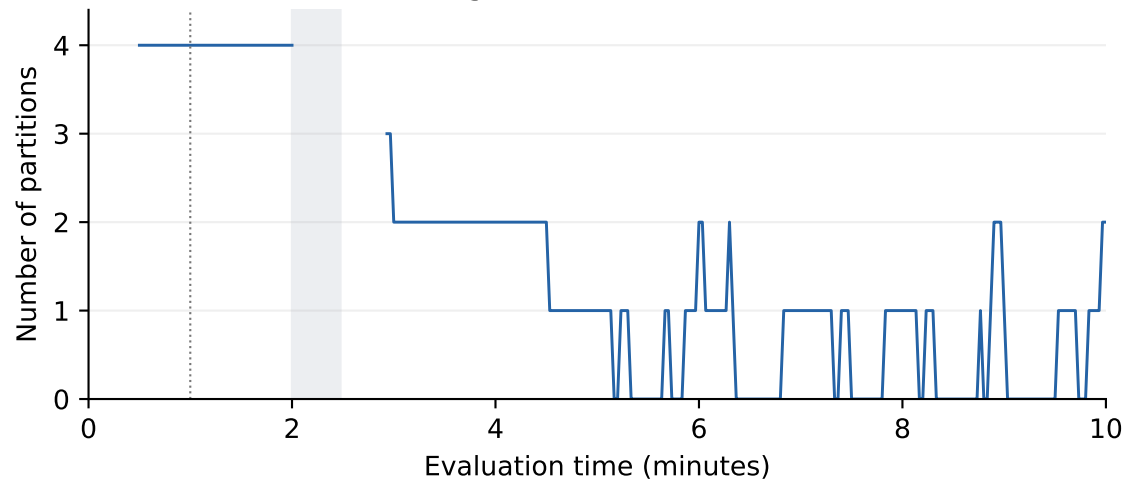
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



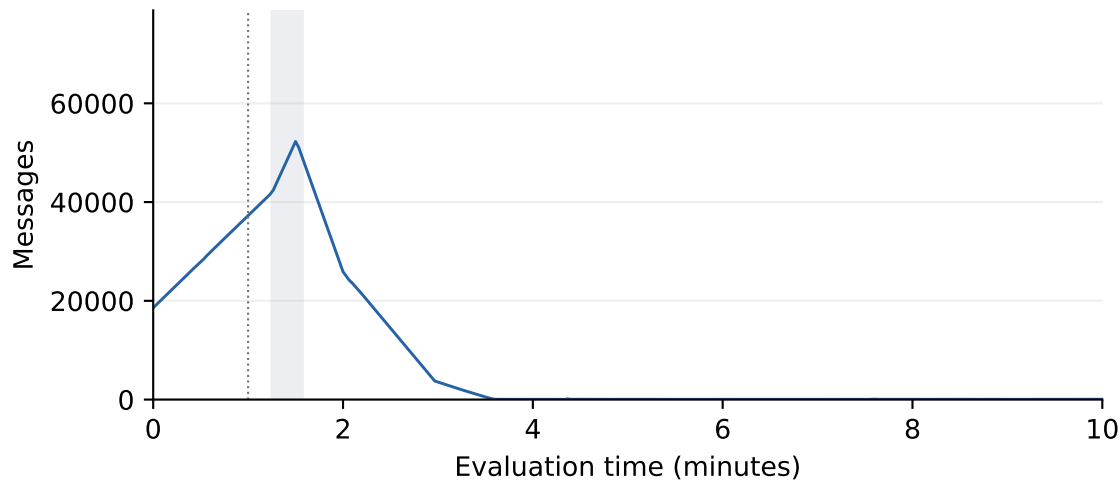
Scale + targeted redistribution - Run 2



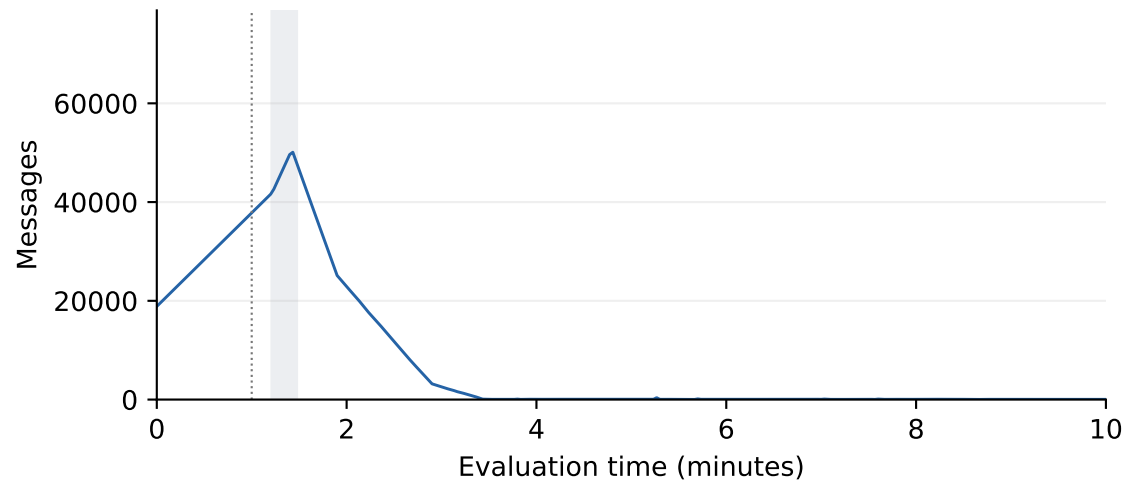
80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.
Hot now, and hot in at least 80% of the last 15 valid snapshots; lag > 10 and > mean + 1 population SD.

Acknowledged but unfinished work - 700 messages/s

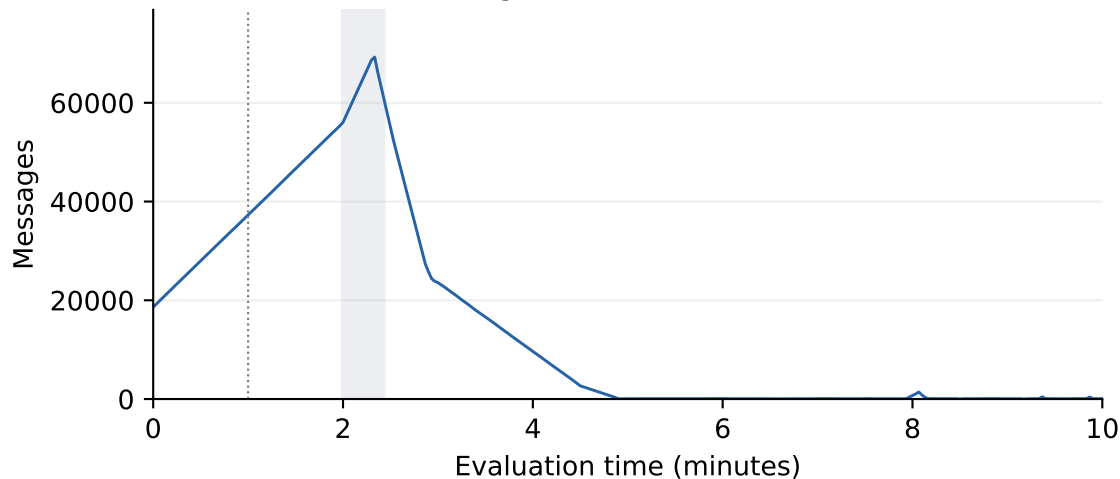
Redistribute within 3 - Run 1



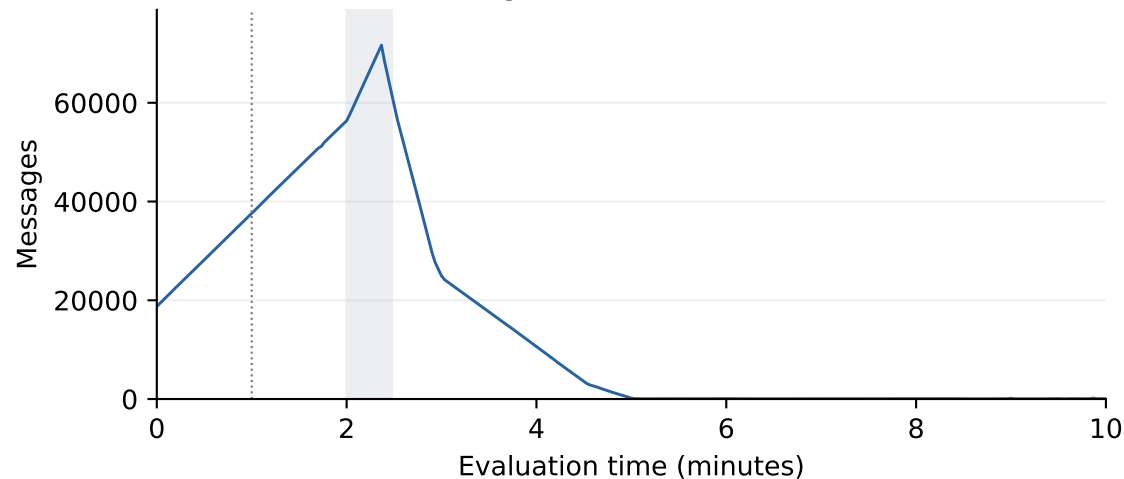
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2

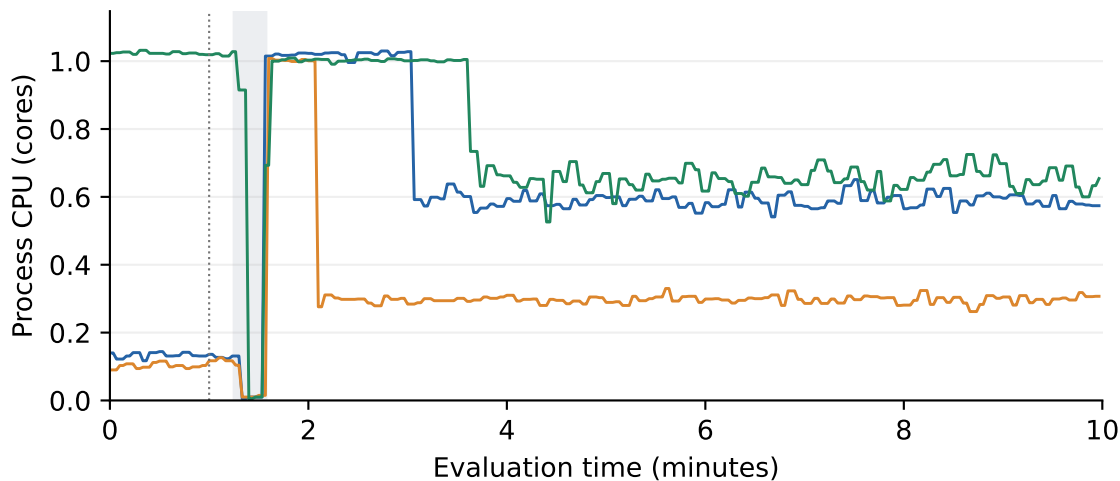


80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.
Distinct acknowledgment/completion events, including warm-up; this is not broker offset lag.

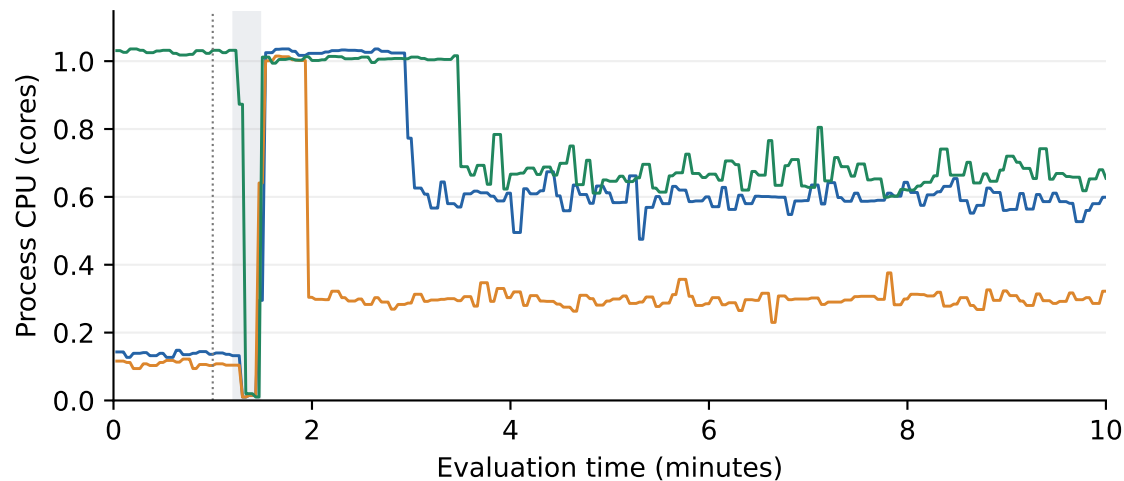
Consumer CPU usage - 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

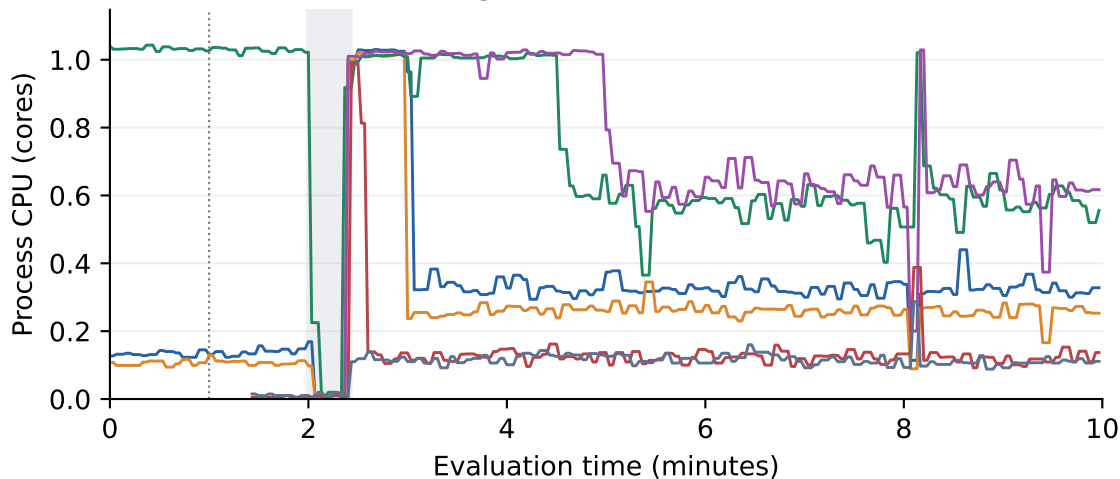
Redistribute within 3 - Run 1



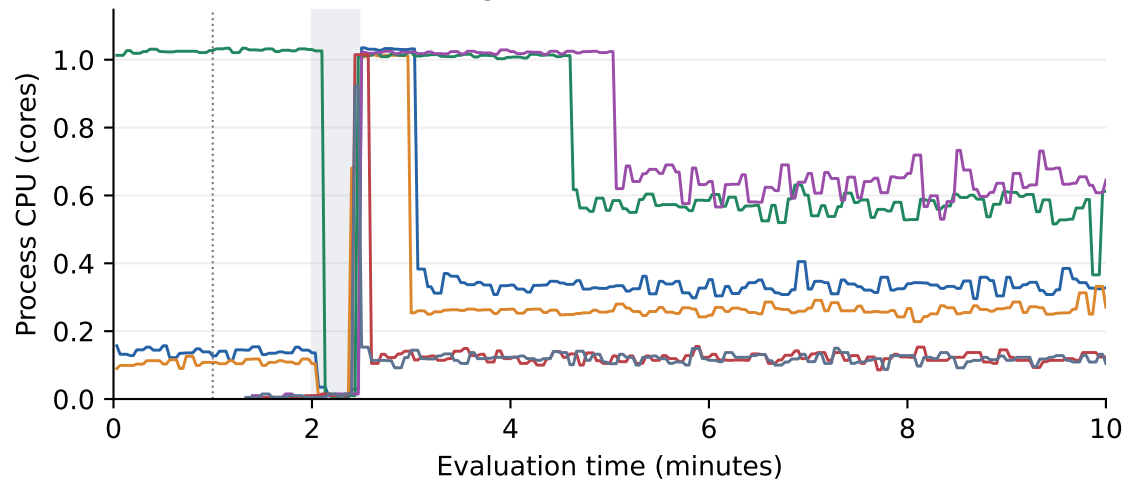
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2



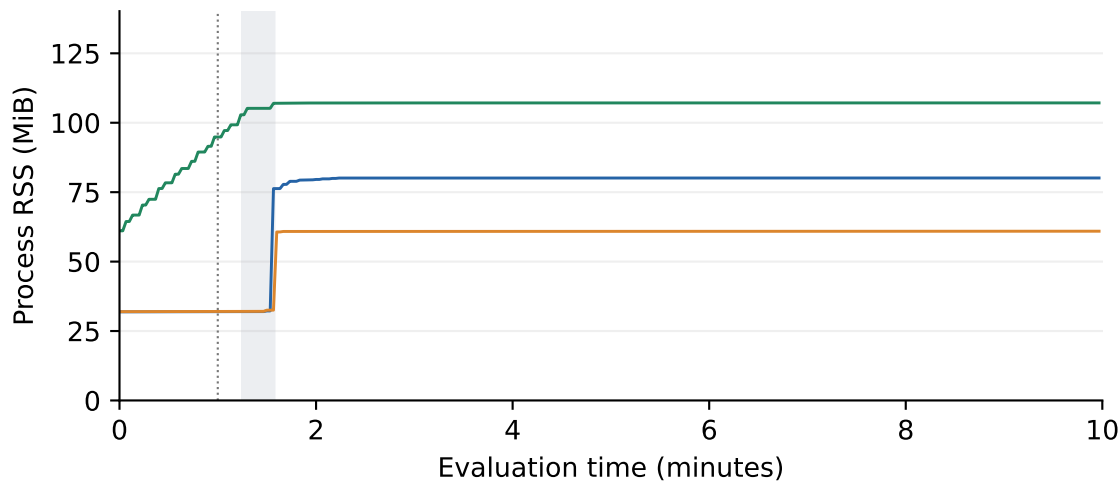
80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.

Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

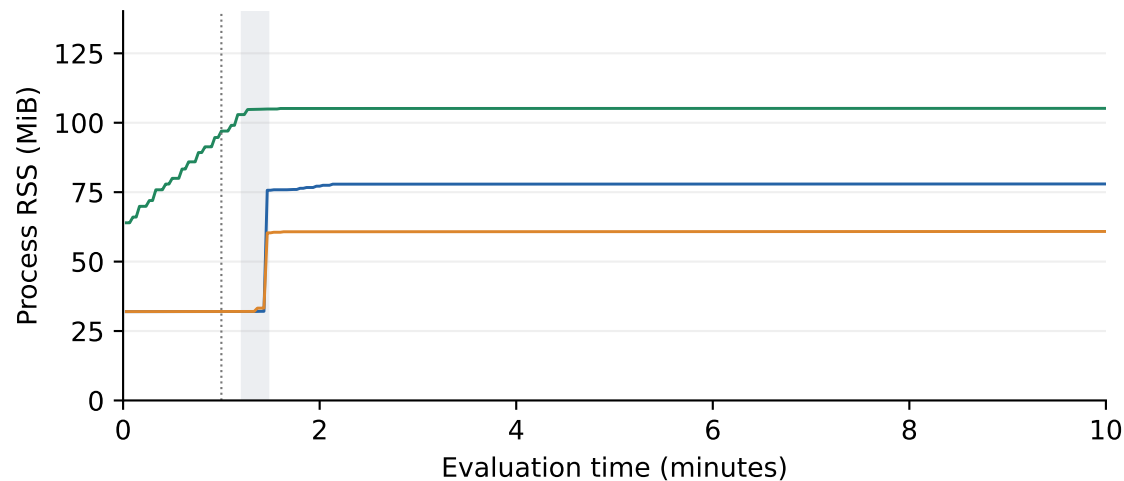
Consumer memory usage - 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

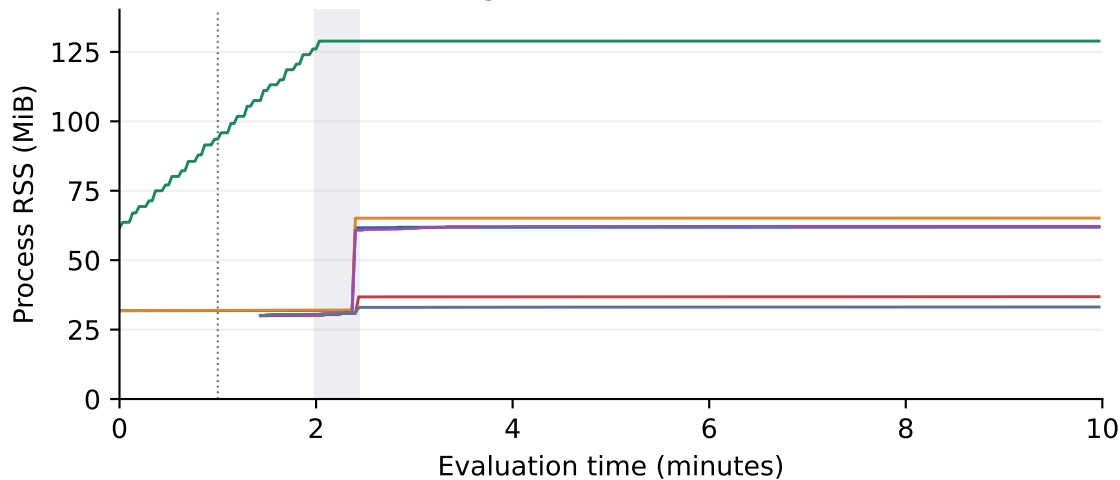
Redistribute within 3 - Run 1



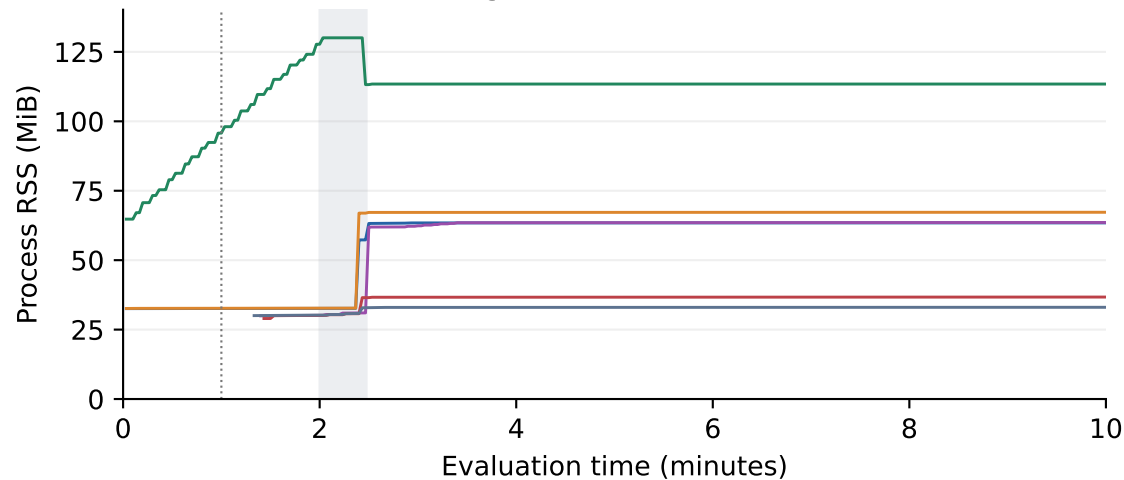
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2

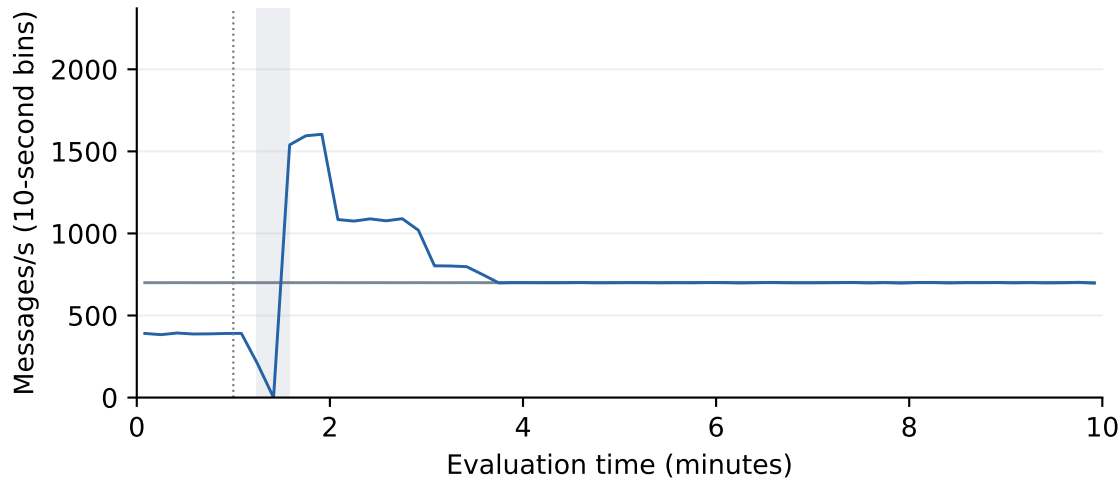


80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

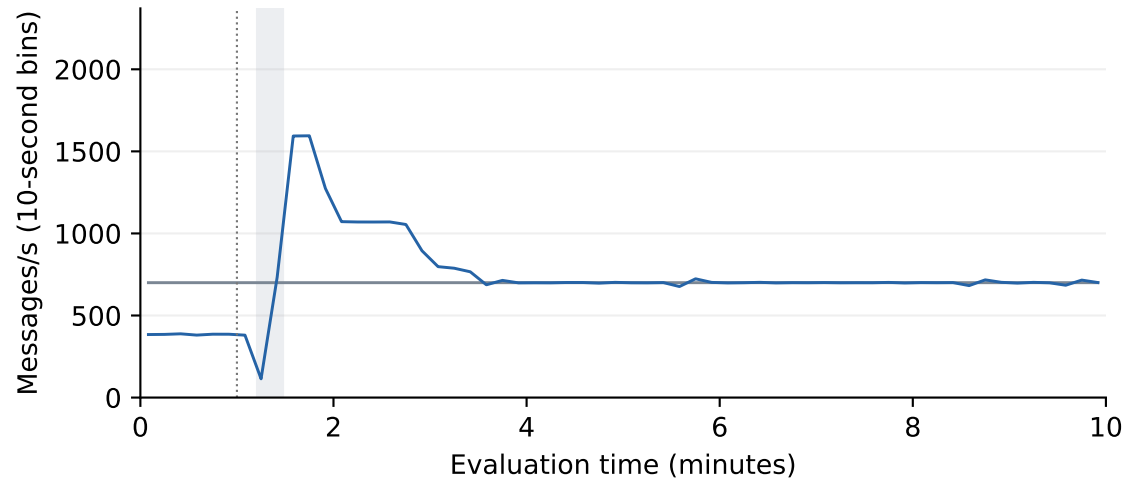
Input and completion throughput - 700 messages/s

Acknowledged input Unique completions

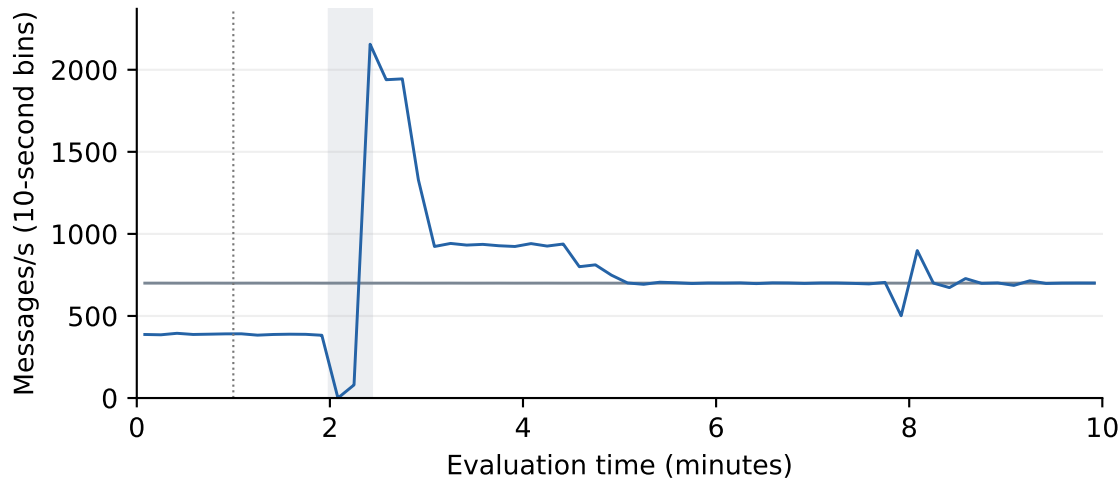
Redistribute within 3 - Run 1



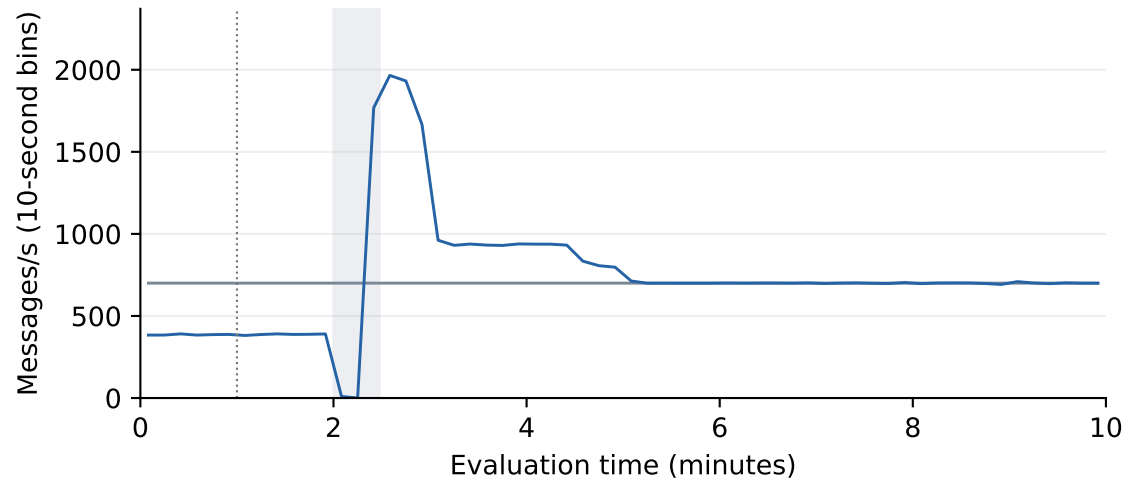
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



Scale + targeted redistribution - Run 2



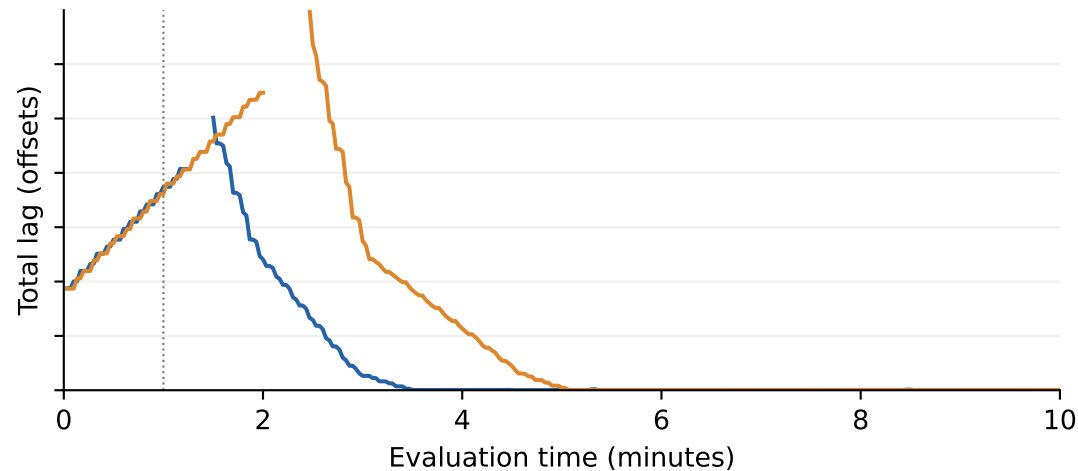
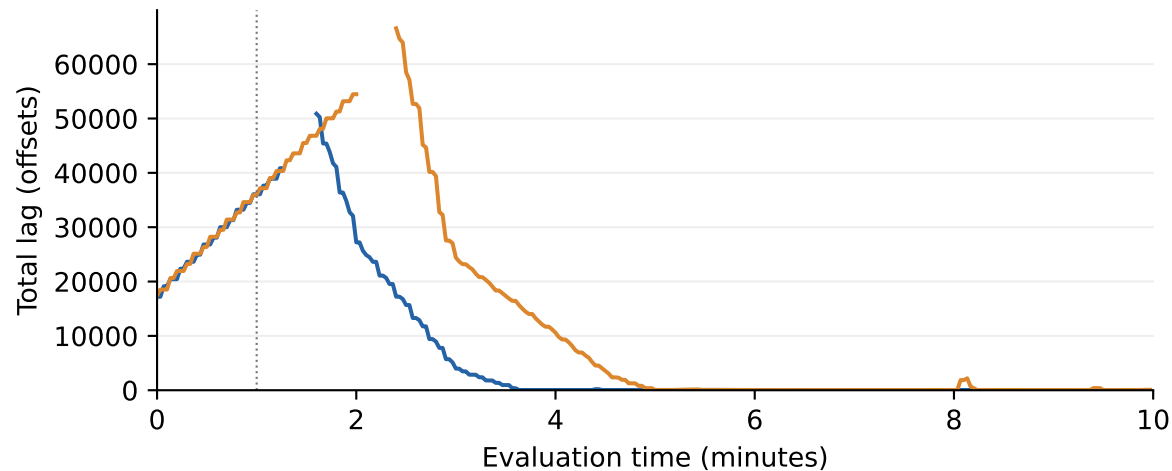
80% of input targets 4 of 60 partitions. Phases: 1 min warm-up + 10 min evaluation + 2 min drain. Gaps remain unavailable.
Completions include warm-up records finishing during evaluation.

Responses from the same concentrated ownership - 700 messages/s

— Redistribute within 3 — Scale + targeted redistribution

Run 1

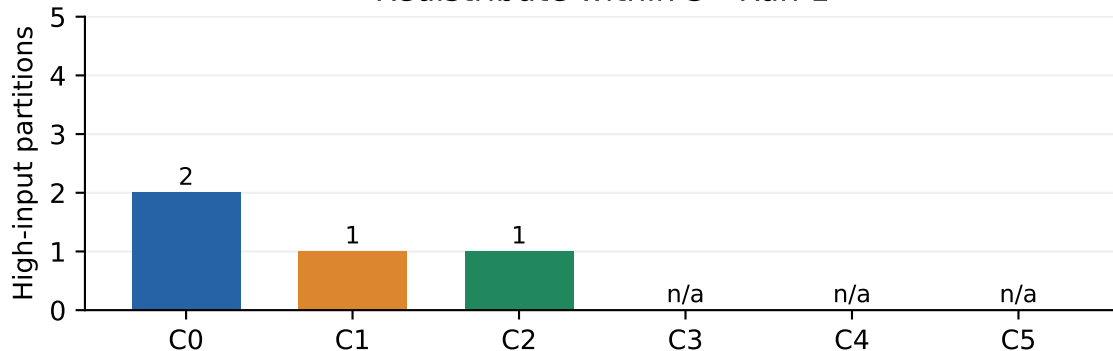
Run 2



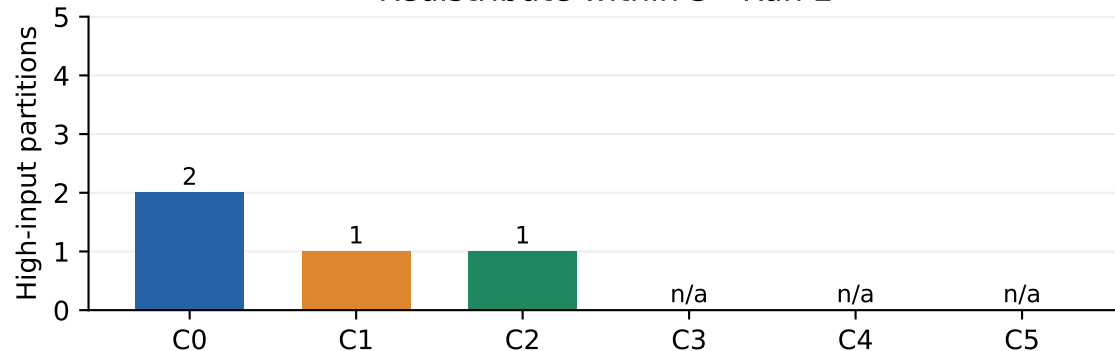
Each curve is a separate 13-minute trial; the 10-minute evaluation is shown. Dotted line: scheduled response. Missing observations and ownership changes break curves. Warm-up work remains queued.

Ownership of the configured high-input partitions at evaluation end

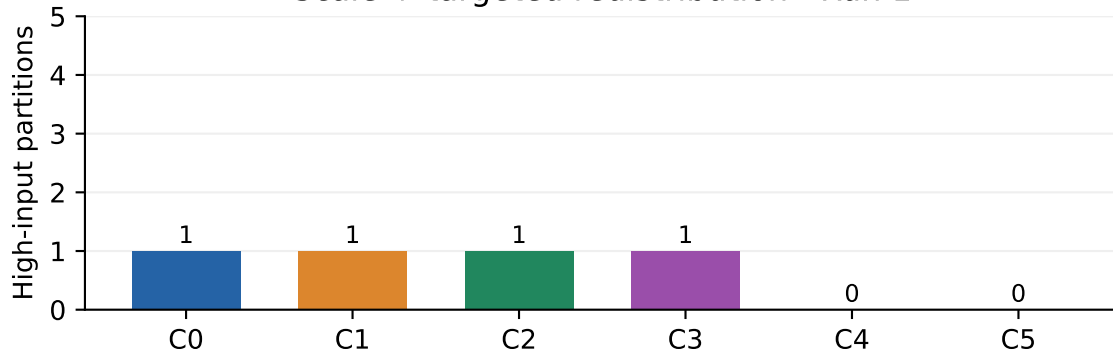
Redistribute within 3 - Run 1



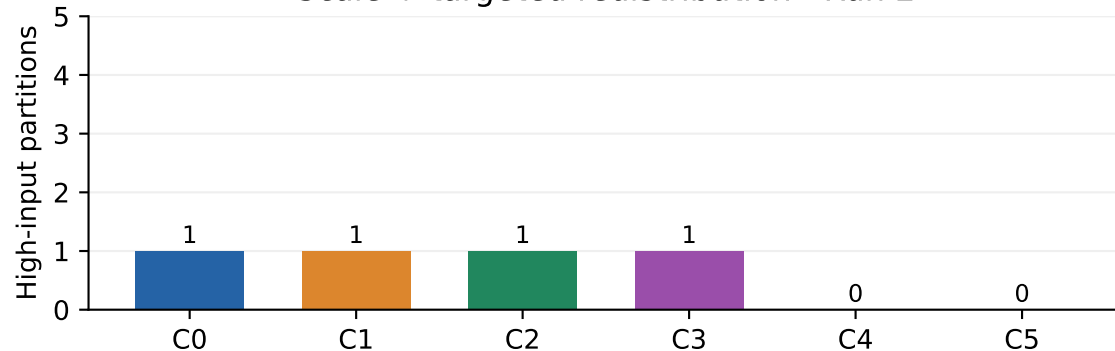
Redistribute within 3 - Run 2



Scale + targeted redistribution - Run 1



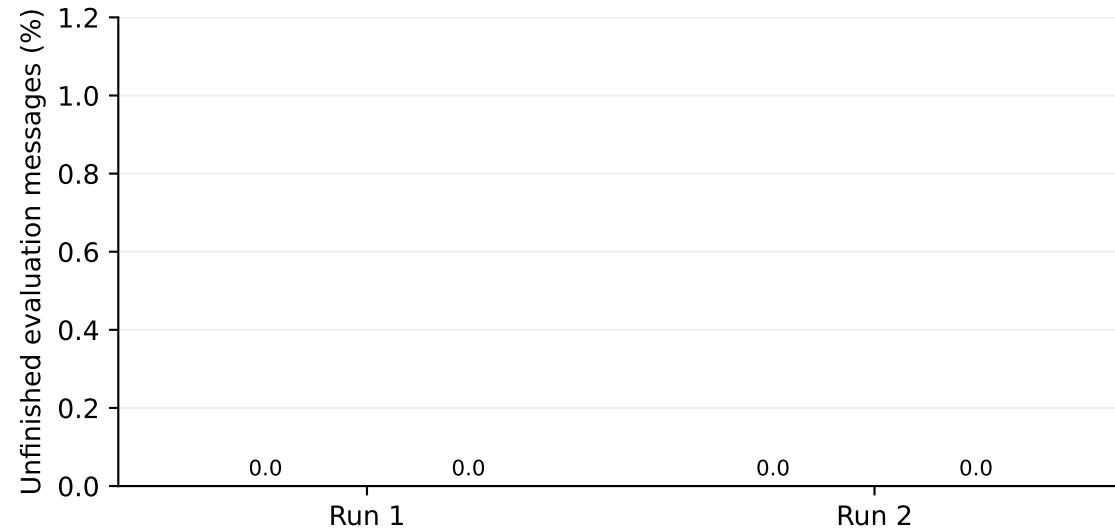
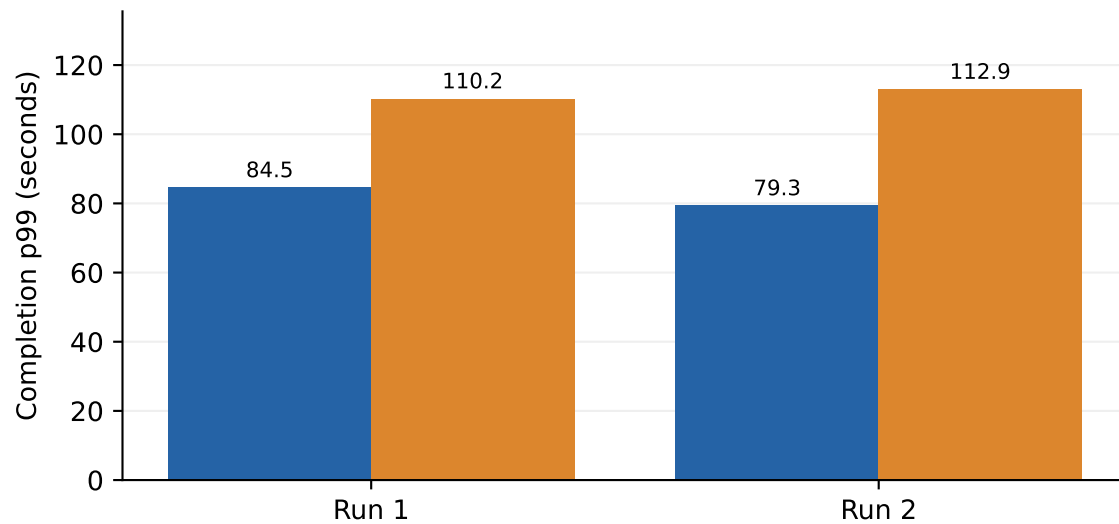
Scale + targeted redistribution - Run 2



All trials started with 4 high-input partitions on Consumer 2. C0-C5 identify consumers. Counts use the last complete valid ownership snapshot; n/a means no partition ownership in that snapshot. These are the configured input targets, not the set classified as persistently hot from lag.

Whole-run completion latency and unfinished work

■ Redistribute within 3 ■ Scale + targeted redistribution



P99 describes completed evaluation messages; unfinished work uses the fixed drain cutoff.

Each trial: 1 min warm-up + 10 min evaluation + 2 min drain. These are separate trials, not within-run before/after p99.